The angle between zero point is the angle of phase difference Θ. It is generally measured in degrees or radians. The quantity which passes through its zero point earlier is said to be leading while the other is said to be lagging.

5.3 ADVANTAGES AND DISADVANTAGES OF AC SUPPLY

Advantages

1) It is easy to transmit alternating current from one place to another place.
2) High voltage can be generated easily.
3) The cost of AC equipment is low.
4) It is possible to convert into DC.
5) Step down, step up voltage can be easily done by transformer.
6) A.C Motors are cheap.

Disadvantages

1) A.C cannot be stored in Batteries.
2) Compared to DC, A.C circuit should have good insulation because of high output voltage which causes electric shock.
3) The voltage drop is occurred due to high starting current in AC.
4) The speed of the AC motor depends upon the frequency.
5) In Inductive load, power factor will be low.

5.4 R-L-C SERIES CIRCUITS

In this circuit, the three (R,L,C) components are all in series with the Voltage source.

5.4.1. AC circuits with pure resistance

A circuit having only resistance and without inductance and capacitance is called pure resistance circuit

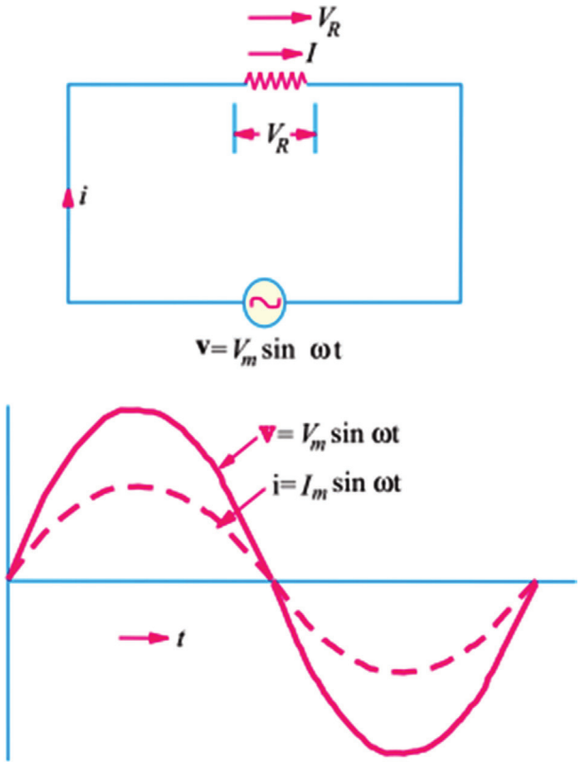


Fig 5.7 *AC through Resistance*

Let

The value of resistance is R

The value of current is I

The value of Electro motive force is E

Then

$$\text{Current (I)} = \frac{\text{Electromotive Force (E)}}{\text{Resistance (R)}}$$

$$I = \frac{E}{R}$$

Power = Current × EMF

In this circuit power factor is unity, because the angle between current and voltage is zero (cosØ = 1).

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